

Tuning block copolymer structural information by adjusting salt concentration in liquid chromatography at critical conditions coupled with electrospray tandem mass spectrometry

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Received 12 January 2009; Revised 12 March 2009; Accepted 13 March 2009

Different cationic adducts of poly(ethylene oxide)/polystyrene block co-oligomers could be produced by adjusting the salt concentration in the mobile phase using a coupling between liquid chromatography at critical conditions and electrospray ionization mass spectrometry. Formation of doubly lithiated adducts was observed at high LiCl concentration (1 mM) while lowering the salt concentration down to 0.1 mM allowed co-oligomers to be ionized with both a proton and a lithium. The fragmentation pathways observed to occur upon collision-induced dissociation of ionized copolymers were shown to be highly dependent on the nature of the cationic adducts. As a result, complementary structural information could be reached by performing MS/MS experiments on different ionic forms of the same co-oligomer molecule. On one hand, release of the nitroxide end-group as a radical from $[M+2Li]^{2+}$ was followed by a complete depolymerization of the polystyrene block, allowing both this end-group and the polystyrene segment size to be determined. On the other hand, $[M+H+Li]^{2+}$ precursor ions mainly dissociated via reactions involving bond cleavages within the nitroxide moiety, yielding useful structural information on this end-group. Copyright © 2009 John Wiley & Sons, Ltd.

Due to their highly ordered nanostructures, block copolymers have become a central component of numerous nanotechnologies.^{1–3} Copolymers containing poly(ethylene oxide) (PEO) as the hydrophilic segment are industrially significant due to their numerous biological and chemical properties.⁴ In particular, combinations of PEO with polystyrene (PS) in a PS-*b*-PEO diblock are used in many industrial applications such as polymeric surfactants, compatibilizers in polymer blending, dispersions, stabilizers and templates for the preparation of inorganic nanoparticles.^{5,6} The self-assembly ordered nanostructures of PEO-*b*-PS has given rise to novel applications such as the preparation of mesoporous silica films with different pore sizes⁷ and electrolyte for rechargeable batteries.²

The unambiguous characterization of block copolymer materials is important since the performance of these materials is highly dependent on structurally related parameters.

The combination of liquid chromatography at critical conditions (LCCC) with mass spectrometry (MS) has been reported to be a powerful tool in such characterizations.^{8–13} At critical conditions, effects due to adsorption and exclusion mechanisms compensate each other and the retention of homopolymer molecules becomes independent of their molecular weight.^{14–20} In addition, MS detection allows the size of the repeat units as well as the end-group masses, the average molecular weights and the degree of polymerization to be determined.²¹ Moreover, as long as a cationizing agent is added to the mobile phase, coupling LCCC on-line with MS detection allows a very straightforward data analysis compared with the complex mass spectra obtained using sample infusion, with both co-monomer blocks displaying their own distribution. For example, performing LCCC/MS for a poly(ethylene oxide)/polystyrene block copolymer (PEO-*b*-PS) using the critical conditions established for PEO,¹³ each mass spectrum consists of signals from co-oligomeric species of a given PS block length; i.e., typically a PEO-like distribution, from which polymer structural information is readily accessible. In these conditions, co-oligomer molecules were mainly separated according to the PS block size, by making the PEO block chromatographically 'invisible'.^{22–30}

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Contract/grant sponsor: French Research Agency; contract/grant number: ANR-06-JCJC-0112.

The presence of a lithium salt in the mobile phase allowed the formation of lithiated adducts during the electrospray ionization (ESI) process while preventing any dilution effects due to post-column salt addition. The type and concentration of the cationizing agent are key parameters for the optimal production of ionic adducts in ESI. The present study shows that, as expected, the salt concentration in the mobile phase has an influence on the ionization yield of PEO-*b*-PS co-oligomers but also, and more interestingly, on the nature of the ions produced during the ESI process. Indeed, at high LiCl concentration (1 mM), doubly charged co-oligomer adducts were the main ionic species produced during electrospray while molecules adducted with both a proton and a lithium cation were formed when the salt concentration in the mobile phase was decreased. As the microstructural characterization of a polymer requires oligomer ions to be fragmented,²¹ the behavior of the two ionic forms of PEO-*b*-PS co-oligomers during collision-induced dissociation (CID) experiments was investigated to check whether complementary tandem mass spectrometric (MS/MS) information could be obtained.

EXPERIMENTAL

Chemicals

Methanol (MeOH) and HPLC grade water were purchased from SDS (Peypin, France) and used without further purification. The salt used to promote electrospray ionization was LiCl from Sigma-Aldrich (St. Louis, MO, USA), used at concentrations ranging from 0.1 to 1 mM in the mobile phase. Hydroxyl-terminated poly(ethylene glycol) standards (PSS, Mainz, Germany), ranging from $M_w = 200$ to 2000 g mol^{-1} , were dissolved in the eluent mixture at a concentration of $100 \mu\text{g mL}^{-1}$. All reaction steps for the preparation of *N*-*tert*-butyl-*N*-[1-(diethylphosphono)-2,2-dimethylpropyl]nitroxide (SG1)-capped PEO-*b*-PS (3) were performed under an atmosphere of dry nitrogen. Poly(ethylene glycol) methyl ether (PEO-Me) with $M_w = 350 \text{ g mol}^{-1}$, styrene (99%), acryloyl chloride (96%) and triethylamine (TEA, 99%) were all purchased from Sigma-Aldrich and were used as received without further purification. The SG1-based alkoxyamine derived from methacrylic acid, also called MAMA-SG1 (2-methyl-2-[*N*-*tert*-butyl-*N*-(1-diethoxyphosphoryl)-2,2-dimethylpropyl] aminoxy] propanoic acid), was kindly supplied by Arkema (Lacq, France). The SG1-capped PEO-*b*-PS was dissolved in the eluent at a concentration of 1 mg mL^{-1} .

SG1-capped PEO-*b*-PS copolymer synthesis

The synthesis has been described in detail in a previous paper.³¹ Briefly, the synthesis of the SG1-capped PEO-*b*-PS diblock copolymer (3) involves three steps: the preparation of PEO-acrylate (1), the intermolecular radical addition of MAMA-SG1 alkoxyamine onto 1 and the polymerization of styrene using PEO-MAMASG1 (2) as an initiator. PEO-acrylate (1) was obtained by esterification of PEO-Me ($M_n = 350 \text{ g mol}^{-1}$) with an acryloyl chloride. Addition of the MAMA-SG1 alkoxyamine to the PEO-acrylate (1) occurs

via thermal homolysis of the C–ON bond leading to the formation of the two radicals: the transitory alkyl radical MAMA and the stable SG1 nitroxide. The PEO-MAMASG1 (2) was then used to initiate the bulk polymerization of styrene at 105°C for 150 min under inert atmosphere. The SG1-capped PEO-*b*-PS diblock copolymer (3) was precipitated in cold *n*-hexane, rinsed with diethyl ether and dried under vacuum at room temperature. Styrene conversion was determined by ^1H NMR spectroscopy ($M_n(\text{PS}) = 300 \text{ g mol}^{-1}$).

Liquid chromatography at critical conditions

LCCC experiments were performed with an Agilent 1100 series system (Agilent Technology, Santa Clara, CA, USA). The chromatographic column (octadecyl-silica Equisil, $150 \times 4 \text{ mm}$, pore diameter: $100 \text{ \AA}$, particle size: $5 \mu\text{m}$), was purchased from C.I.L. Cluzeau (Sainte Foy La Grande, France) and was maintained at a constant temperature of 25°C . For all experiments, the injection volume was $20 \mu\text{L}$ and the flow rate was set to 1 mL min^{-1} . Prior to entering the ESI source, the chromatographic effluent was split down to $50 \mu\text{L min}^{-1}$ using a zero-dead-volume tee connector. The critical conditions were established for hydroxyl-terminated PEG standards, ranging from $M_w = 200$ to 2000 g mol^{-1} , using 1 mM LiCl concentration. The critical composition of the MeOH/water eluent was determined to be 88.8:11.2 (% v/v) and this allows elution of all the PEG chains at the same retention time ($t_r = 2.33 \text{ min}$) independently of their molecular mass.¹³ LiCl concentration in the range 0.1–1 mM led to the same critical conditions being obtained.

Fig 1

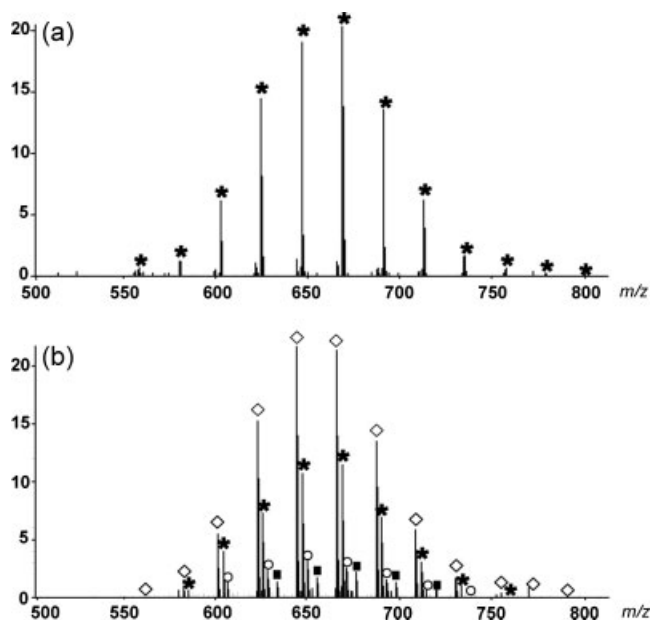


Figure 1. Mass spectra extracted from the chromatographic peak corresponding to $m = 4$ at critical conditions of PEO using LiCl concentration of (a) 1 mM and (b) 0.1 mM LiCl. Detected ionic species are $[\text{M} + \text{H} + \text{Li}]^{2+}$ ($\diamond$), $[\text{M} + 2\text{Li}]^{2+}$ (*), $[\text{M} + \text{H} + \text{NH}_4]^{2+}$ (o) and $[\text{M} + \text{Na} + \text{Li}]^{2+}$ ($\blacksquare$).

Mass spectrometry

Mass spectrometry was performed with a QStar Elite mass spectrometer (Applied Biosystems Sciex, Concord, ON, Canada) equipped with an electrospray ionization (ESI) source, operated in the positive ion mode. The capillary voltage was set at 5500 V and the cone voltage at 80 V. In this hybrid instrument, ions were measured using an orthogonal acceleration time-of-flight (TOF) mass analyzer. A quadrupole analyzer was used for the selection of precursor ions to be further submitted to collision-induced dissociation (CID) in MS/MS experiments. Nitrogen was used as the nebulizing gas (at a pressure of 20 psi) and the curtain gas (20 psi). Analyst software version 2.1 (Applied Biosystems Sciex) was used for instrument control, data acquisition and data processing for both chromatography and mass spectrometry. Calibration of the TOF analyzer during MS/MS experiments was performed using the accurate masses of the precursor ions.

RESULTS AND DISCUSSION

LCCC/MS of the SG1-capped PEO-*b*-PS copolymer

The retention of the SG1-capped PEO-*b*-PS copolymer was examined at the critical conditions of the PEO block in order to make the PEO segment chromatographically 'invisible' and to separate the diblock copolymer according to the size of the PS block. These critical conditions were determined using an eluent containing the electrospray cationizing agent (i.e. LiCl). The optimization of the experimental conditions as well as a detailed description of each chromatographic peak have previously been reported.¹³ Briefly, up to seven peaks were observed, each of which corresponded to PEO_n-*b*-PS_m molecules with a given PS block size (*m* varying from 1 to 7). The ESI mass spectra extracted from each co-oligomer peak showed unique PEO-like distributions of doubly lithiated

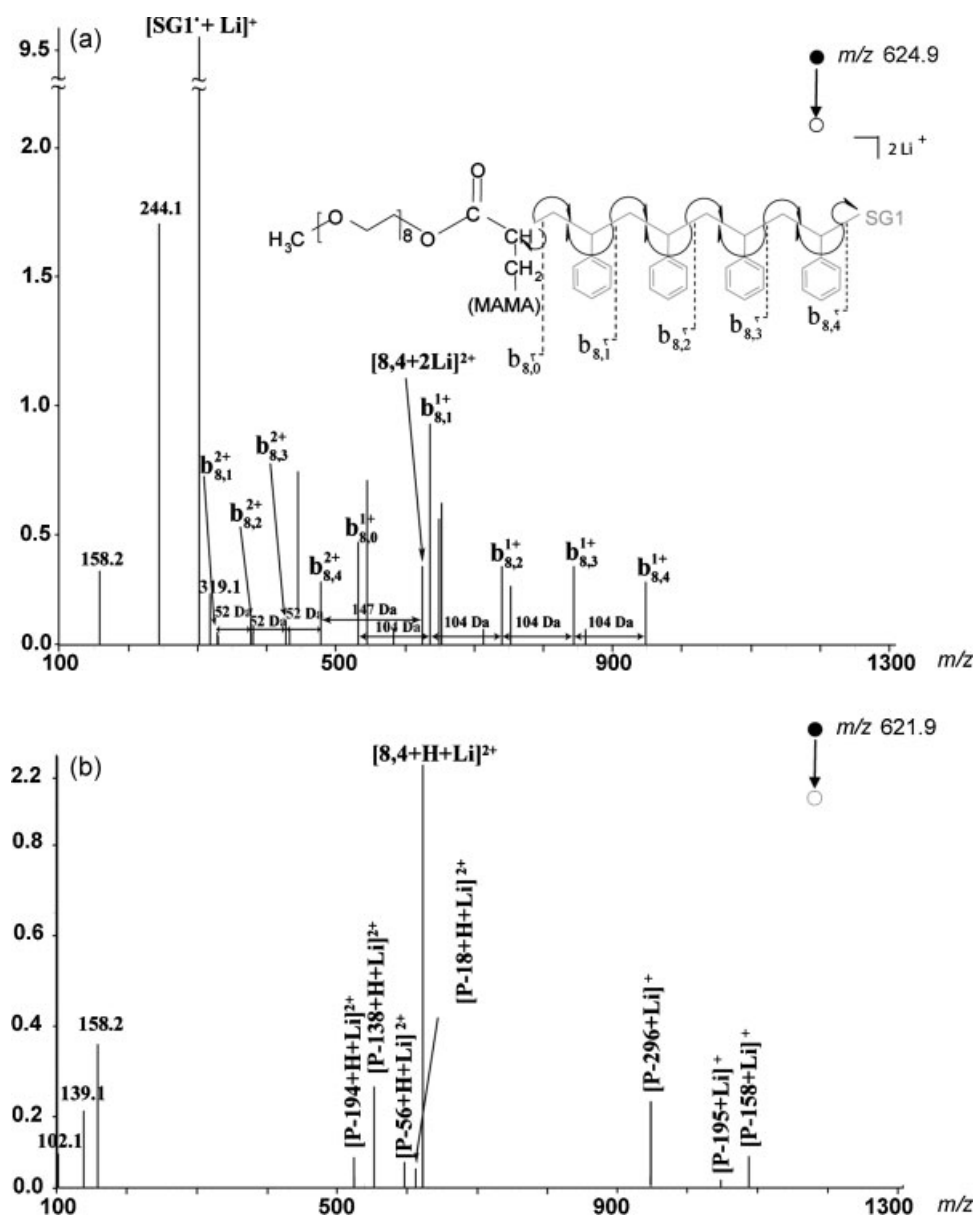
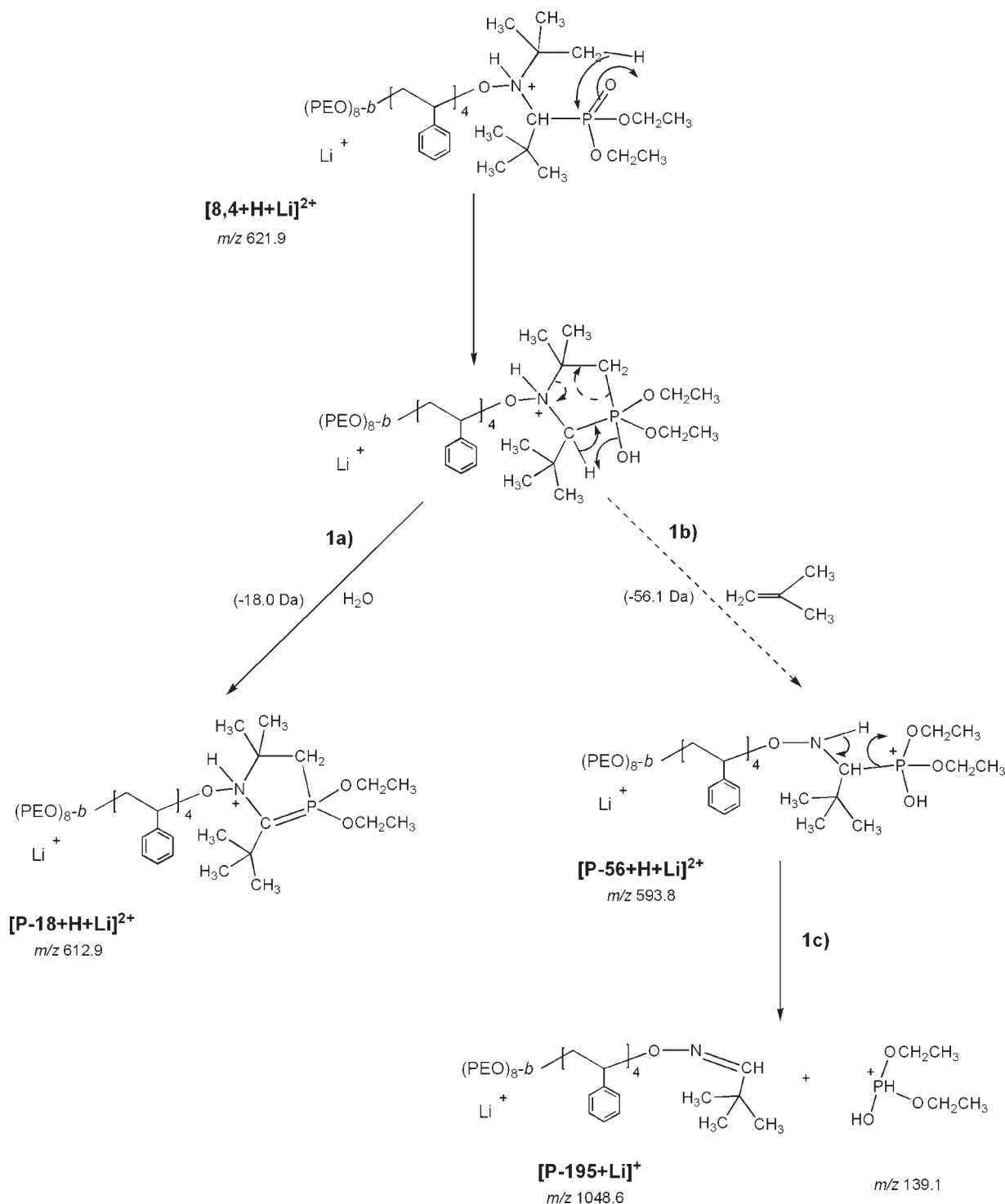


Figure 2. ESI-MS/MS spectra of the SG1-capped PEO₈-*b*-PS₄ co-oligomer doubly cationized (a) with Li⁺, [M+2Li]²⁺, at *m/z* 624.9 (using 1 mM LiCl) and (b) with both a proton and a lithium, [M+H+Li]²⁺, at *m/z* 621.9 (using 0.1 mM LiCl). The CID experiments were performed at 20 eV collision energy (laboratory frame).

molecules, as indicated both by the difference of 22 m/z units measured between successive ions and by the ion isotope pattern.

The signal-to-noise ratios in the ESI mass spectra were observed to decrease as the size of the PS block increased. The salt concentration in the mobile phase is a key parameter in ESI and can be varied to obtain optimal ionization yields. Increasing the LiCl concentration above 1 mM did not provide any improvement of the signal intensity. In contrast, the abundance of co-oligomer ions in the mass spectra was

enhanced on decreasing the LiCl salt concentration in the eluent. The chromatographic signal is about 1.4 times higher using 0.1 mM LiCl than when using 1 mM. It should also be noted that variation of the salt concentration in the mobile phase has no influence on the critical conditions, as first established for 1 mM of LiCl. Interestingly, decreasing the salt concentration from 1 mM to 0.1 mM results in the production of different co-oligomer adducts. Figure 1 shows the LCCC/MS spectra of co-oligomers with four styrene units ($m = 4$) at two different salt concentrations in the



Scheme 1. Dissociation mechanisms proposed to account for the formation of (a) m/z 612.9 and (b) m/z 593.8 from m/z 621.9, as well as (c) consecutive dissociation of m/z 593.8 to produce m/z 1045.6 and 139.1.

mobile phase. In contrast to the unique doubly lithiated distribution of PEO-like molecules observed with 1 mM LiCl (Fig. 1(a)), four co-oligomeric envelopes were observed using 0.1 mM LiCl. In Fig. 1(b), the $[M+2Li]^{2+}$ ions are still detected but the most intense distribution arises from $[M+H+Li]^{2+}$ ions. In addition, very low abundance $[M+H+NH_4]^{2+}$ and $[M+Na+Li]^{2+}$ co-oligomer adducts are observed, whose formation was attributed to residual NH_4^+ and Na^+ in the ESI source, respectively. Preferential formation of $[M+H+Li]^{2+}$ co-oligomer ions over the doubly lithiated adducts would reflect that protons efficiently compete with Li^+ for co-oligomer molecule ionization as the LiCl concentration is lowered.

MS/MS experiments were then performed on the most abundant ions in the ESI mass spectra, namely $[M+2Li]^{2+}$ and $[M+H+Li]^{2+}$, to check whether the nature of adducted cations has an influence on the dissociation behavior of the co-oligomer ions.

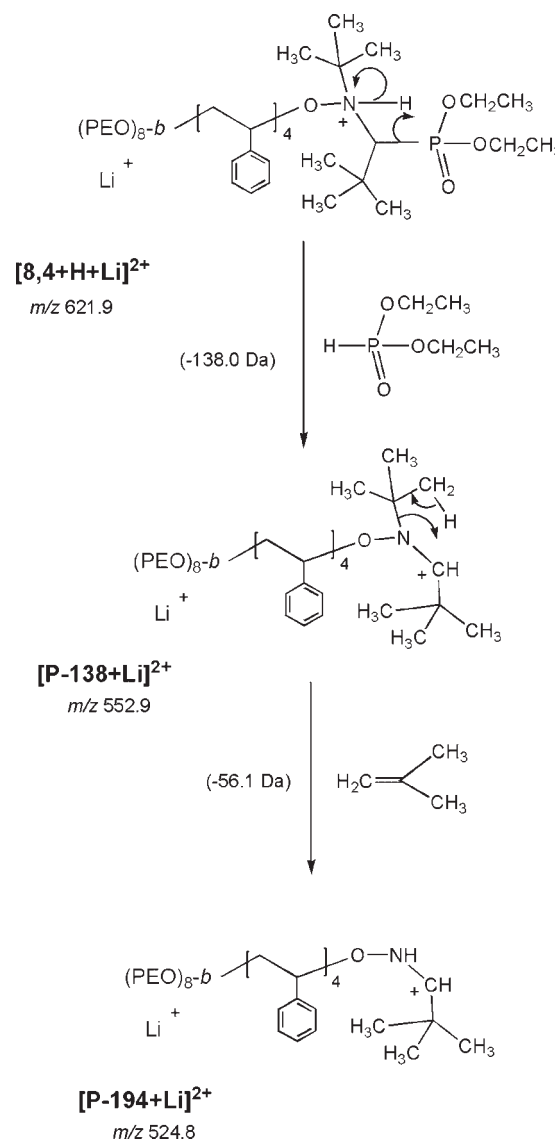
LC/MS/MS of the SG1-capped PEO-*b*-PS copolymer

The abundance of the $[M+2Li]^{2+}$ and $[M+H+Li]^{2+}$ ions in the MS spectra (using a 1 mM and a 0.1 mM LiCl concentration, respectively) was sufficient to allow MS/MS experiments to be performed. Figure 2(a) shows the MS/MS spectrum of the doubly lithiated SG1-capped PEO₈-*b*-PS₄ molecule, noted $[8,4+2Li]^{2+}$. Dissociation of doubly lithiated adducts could be accounted for using mechanisms previously established for singly lithiated co-oligomers.³¹ This result is consistent with data from a recent study which showed that the charge state of the ion has no influence on its fragmentation behaviour.¹³ The main dissociation route of this m/z 624.9 ion was observed to proceed via the homolytic cleavage of the C–ON bond between the last styrene unit and the SG1 end-group, which undergoes reversible homolysis during nitroxide-mediated polymerization. This reaction produces $[SG1^{\cdot}+Li]^+$ (m/z 301.1892, error: –2.3 ppm) and $b_{8,4}^+$ (m/z 948.5608, error: +3.9 ppm) as complementary ions, or $b_{8,4}^{2+}$ (m/z 477.7867, error: +0.8 ppm), depending on the location of Li^+ cations at the time of the dissociation. Details of the reaction mechanisms as well as a definition of the employed ion nomenclature have been described elsewhere.^{13,31–33} This homolytic bond cleavage induces a depolymerization process of the PS block from the so-formed doubly or singly charged radical, as depicted in the inset of Fig. 2(a). These successive eliminations of styrene molecules (104.1 Da neutral losses) allowed a straightforward determination of the PS block size. Although additional fragmentation processes also occurred during the CID of the precursor ion,^{13,31} information obtained during these MS/MS experiments mainly indicates the size of the PS block and the mass of the terminating end-group.

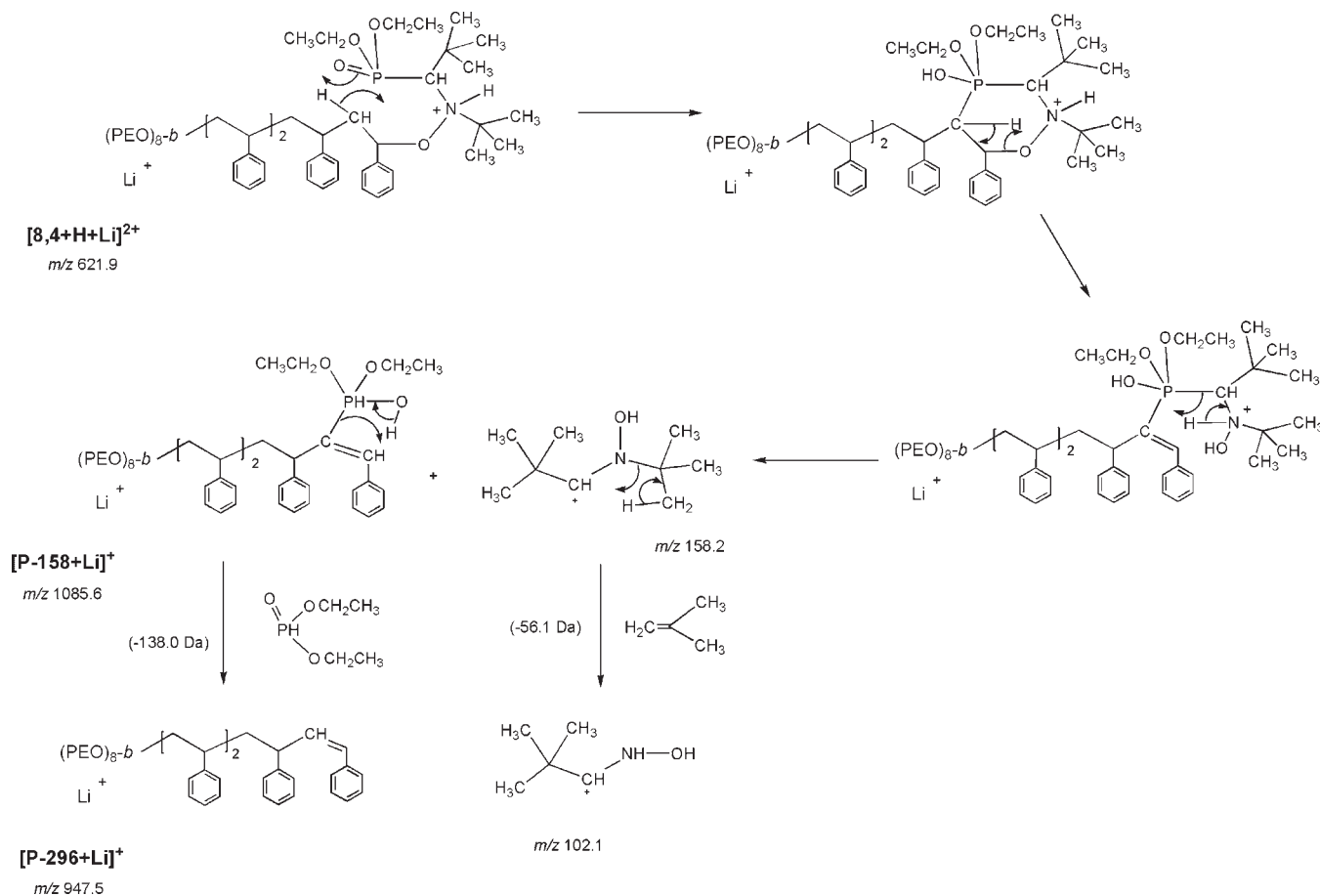
In contrast, CID of $[M+H+Li]^{2+}$ ions gives rise to less complex MS/MS spectra, as illustrated in Fig. 2(b) for the SG1-capped PEO₈-*b*-PS₄ molecule cationized with both a proton and a lithium ion. First, the homolytic cleavage of the C–ON bond, previously observed from doubly lithiated precursor ions, and followed by the complete depolymerization of the PS block, is no longer observed. As

demonstrated in a recent study, the C–ON bond dissociation energy increases by about 25 kcal mol^{–1} upon protonation of the nitrogen of the SG1 group.³⁴ The release of the nitroxide radical would thus no longer efficiently compete with alternative dissociation pathways, as indicated by the MS/MS data (Fig. 2(b)).

Loss of a water molecule from the m/z 621.9 precursor ion is indicated by a product ion at m/z 612.9 (m/z 612.8695, error: +1.5 ppm) and would occur according to the mechanism depicted in Scheme 1(a). This dehydration process would imply first the protonation of the P=O group via a proton transfer from one methyl group, yielding a five-membered ring intermediate. A consecutive 1,3-water elimination would then allow the doubly charged $[P-18+H+Li]^{2+}$ ion to be produced. An alternative reaction pathway from the five-membered ring intermediate could also be envisaged. Indeed, a concerted electron migration within this ring would allow the release of a 2-methylprop-1-ene neutral (C_4H_8 , 56.1 Da) and the formation of the $[P-56+H+Li]^{2+}$ ion (m/z 593.8464, error: +6.4 ppm) as indicated in Scheme 1(b). This so-formed $[P-56+H+Li]^{2+}$ ion would further dissociate



Scheme 2. Reaction pathway proposed for the formation of m/z 552.9 and 524.8 from m/z 621.9.



Scheme 3. Dissociation reaction proposed to account for the formation, from the precursor ion at *m/z* 621.9, of complementary product ions *m/z* 1085.6 and 158.2, which would further dissociate to yield *m/z* 947.5 and 102.1, respectively.

via a 1,3-proton transfer from the nitrogen to the phosphorus atom (Scheme 1(c)) allowing the release of two singly charged species, *m/z* 139.1 ion (*m/z* 139.0526, error: +5.1 ppm) and the so-called [P-195+Li]⁺ ion (*m/z* 1048.6242, error: -8.7 ppm).

Another dissociation reaction of the precursor ion would consist of a 1,3-proton transfer from the nitrogen to the phosphorus atom and would allow the release of a neutral diethyl phosphonate (C₄H₁₁O₃P, 138.0 Da) species and the formation of a doubly charged ion, noted [P-138+Li]²⁺ (*m/z* 552.8519, error: +0.6 ppm), as described in Scheme 2. The proposed mechanism is consistent with recently reported theoretical investigations on C-P bond cleavage induced by protonation of α -aminophosphonates.³⁵ No Li⁺ adduct of the diethyl phosphonate is detected, suggesting that the lithium is preferentially located on the co-oligomeric chain, in particular the PEO block.³⁶ A 2-methylprop-1-ene neutral (C₄H₈, 56.1 Da) would be released from this *m/z* 552.9 ion, yielding the doubly charged [P-194+Li]²⁺ ion (*m/z* 524.8207, error: +0.8 ppm), after a 1,3-proton transfer has occurred from a methylene group to the nitrogen atom (Scheme 2).

Accurate mass data indicate that the singly charged ions detected at *m/z* 1085.6 (*m/z* 1085.6002, error: +5.8 ppm) and *m/z* 158.2 (*m/z* 158.1528, error: -6.9 ppm) are complementary ions formed during the same dissociation reaction from the precursor ion. To account for this reaction, a three-step mechanism could be proposed (Scheme 3). A proton transfer from the methylene group of the last styrene unit to the

oxygen atom of the P=O moiety would first give rise to the formation of a six-membered ring. This ring would then be opened via a 1,3-proton transfer allowing the formation of a OH group. Finally, a second 1,3-proton transfer from the nitrogen to the phosphorus atom would produce the *m/z* 158.2 ion and the lithiated adduct, [P-158+Li]⁺, at *m/z* 1085.6. As depicted in Scheme 3, the *m/z* 158.2 ion would further eliminate a 2-methylprop-1-ene neutral to yield the ion at *m/z* 102.1 (*m/z* 102.0919, error: +6.7 ppm), while the [P-158+Li]⁺ ion would release the diethyl phosphonate (C₄H₁₁O₃P, 138.0 Da) moiety to produce the lithiated adduct [P-296+Li]⁺ (*m/z* 947.5477, error: -1.7 ppm).

This MS/MS data analysis reveals that when these PEO-*b*-PS co-oligomers are cationized with both a proton and a lithium, collision-induced dissociations mainly occur in the SG1 end-group, which could thus be characterized in detail.

CONCLUSIONS

This study shows the influence of the salt concentration in the mobile phase using a LCCC/ESI-MS coupling for the characterization of a SG1-terminated PEO-*b*-PS copolymer. Decreasing the LiCl concentration in the eluent allowed a slight improvement of the co-oligomer ESI yield. More interestingly, formation of different co-oligomeric adducts is observed as a function of salt concentration, with the main distributions in the MS spectra arising from [M+2Li]²⁺ and

the $[M+H+Li]^{2+}$ using 1 mM and 0.1 mM LiCl, respectively. The nature of the cations adducted to the co-oligomers was shown to dictate the precursor ion dissociation pathways. On one hand, $[M+2Li]^{2+}$ fragmentation consists of the release of the SG1 end-group followed by a complete depolymerization of the PS block, allowing a straightforward determination of both the terminal end-group and the PS block size. On the other hand, detailed structural information of the SG1 end-group could be obtained during CID of $[M+H+Li]^{2+}$ precursor ions. Indeed, the presence of a nitrogen atom in the co-oligomer allows the protonation of this functionality, further preventing the release of SG1 which then undergoes extensive internal fragmentation. The salt concentration in the mobile phase can then be adjusted according to the structural information being targeted.

Acknowledgements

This work was supported by the French Research Agency (ANR-06-JCJC-0112). L. Charles acknowledges support from Spectropole, the Analytical Facility of Aix-Marseille University, by allowing a special access to the instruments purchased with European Funding (FEDER OBJ2142-3341). T.N.T. Phan acknowledges Arkema for kindly supplying the MAMA-SG1 alkoxyamine.

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